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Apache Hadoop : Creating Wordcount Java Project with Eclipse Part 2 - 2015




K Hong

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Big Data & Hadoop Tutorials

Hadoop 2.6 -
Installing on
Ubuntu 14.04
(Single-Node
Cluster)

Hadoop - Running
MapReduce Job

Hadoop -
Ecosystem

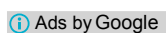
CDH5.3 Install on
four EC2 instances
(1 Name node and
3 Datanodes)
using Cloudera
Manager 5

CDH5 APIs

QuickStart VMs for
CDH 5.3

QuickStart VMs for
CDH 5.3 II - Testing
with wordcount

QuickStart VMs for
CDH 5.3 II - Hive
DB query


[▶ Hadoop Mapreduce](#)
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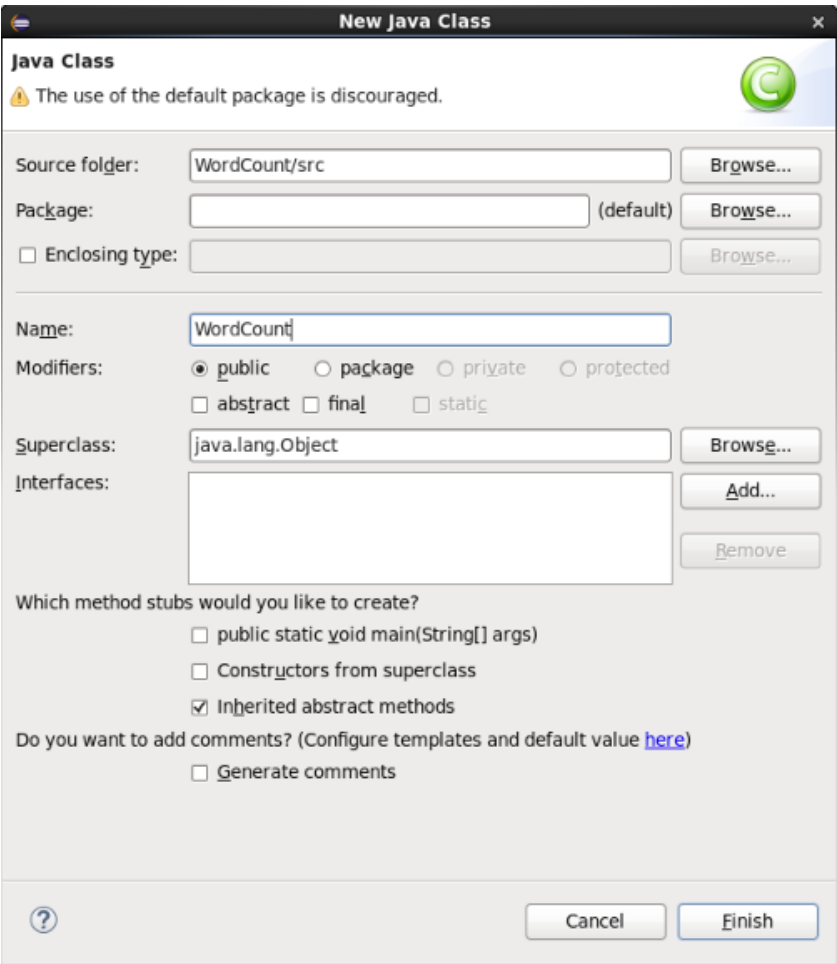
note

In this chapter, we'll continue to create a Wordcount

Java project with Eclipse for Hadoop. For Part 1, please visit [Apache Hadoop : Creating Wordcount Java Project with Eclipse](#). In the previous chapter, we created a WordCount project and got external jars from Hadoop. Now, we'll wrap up our MapReduce work in this chapter.

Creating class files

Right click on source, New > Class:



Click "Finish".

Scheduled start and stop CDH services

Zookeeper & Kafka Install

Zookeeper & Kafka - single node single broker cluster

Zookeeper & Kafka - single node multiple broker cluster

OLTP vs OLAP

Apache Hadoop Tutorial I with CDH - Overview

Apache Hadoop Tutorial II with CDH - MapReduce Word Count

Apache Hadoop Tutorial III with CDH - MapReduce Word Count 2

Apache Hadoop (CDH 5) Hive Introduction

CDH5 - Hive Upgrade to 1.3 to from 1.2

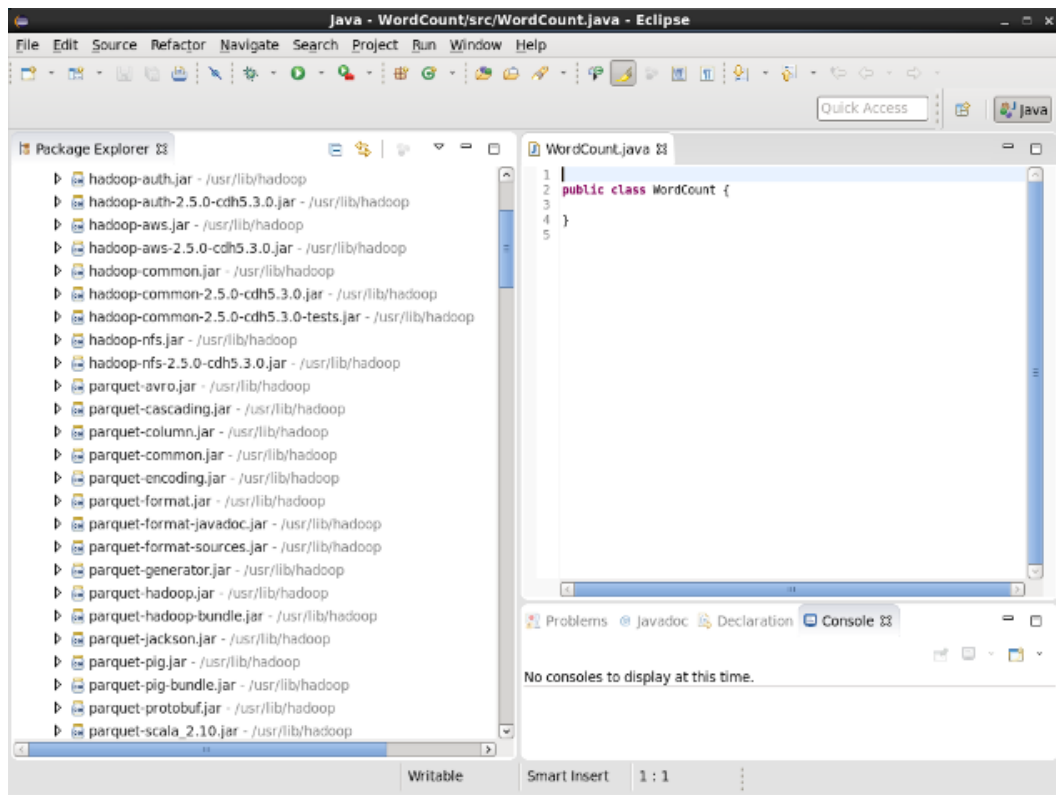
Apache Hadoop : Creating HBase table with HBase shell and HUE

Apache Hadoop : Creating HBase table with Java API

Apache Hadoop : HBase in Pseudo-Distributed mode

Apache Hadoop HBase : Map, Persistent, Sparse, Sorted, Distributed and Multidimensional

Apache Hadoop - Flume with CDH5: a single-node Flume deployment (telnet example)



Eclipse using
Cloudera VM
UnoExample for
CDH5 - local run

We may want to get the code from

<http://wiki.apache.org/hadoop/WordCount>:

Apache Hadoop :
Creating

```
import java.io.IOException;
import java.util.*;

import org.apache.hadoop.fs.Path;
import org.apache.hadoop.conf.*;
import org.apache.hadoop.io.*;
import org.apache.hadoop.mapreduce.*;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.input.TextInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
import org.apache.hadoop.mapreduce.lib.output.TextOutputFormat;

public class WordCount {

    public static class Map extends Mapper {
        private final static IntWritable one = new IntWritable(1);
        private Text word = new Text();

        public void map(LongWritable key, Text value, Context context) throws IOException, InterruptedException {
            String line = value.toString();
            StringTokenizer tokenizer = new StringTokenizer(line);
            while (tokenizer.hasMoreTokens()) {
                word.set(tokenizer.nextToken());
                context.write(word, one);
            }
        }
    }

    public static class Reduce extends Reducer {

        public void reduce(Text key, Iterable values, Context context)
            throws IOException, InterruptedException {
            int sum = 0;
            for (IntWritable val : values) {
                sum += val.get();
            }
            context.write(key, new IntWritable(sum));
        }
    }
}
```

```

    }
}

public static void main(String[] args) throws Exception {
    Configuration conf = new Configuration();

    Job job = new Job(conf, "wordcount");

    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(IntWritable.class);

    job.setMapperClass(Map.class);
    job.setReducerClass(Reduce.class);

    job.setInputFormatClass(TextInputFormat.class);
    job.setOutputFormatClass(TextOutputFormat.class);

    FileInputFormat.addInputPath(job, new Path(args[0]));
    FileOutputFormat.setOutputPath(job, new Path(args[1]));

    job.waitForCompletion(true);
}
}

```

```

WordCount.java
1 import java.io.IOException;
2 import java.util.*;
3
4 import org.apache.hadoop.fs.Path;
5 import org.apache.hadoop.conf.*;
6 import org.apache.hadoop.io.*;
7 import org.apache.hadoop.mapreduce.*;
8 import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
9 import org.apache.hadoop.mapreduce.lib.input.TextInputFormat;
10 import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
11 import org.apache.hadoop.mapreduce.lib.output.TextOutputFormat;
12
13 public class WordCount {
14
15     public static class Map extends Mapper<LongWritable, Text, Text, IntWritable> {
16         private final static IntWritable one = new IntWritable(1);
17         private Text word = new Text();
18
19         public void map(LongWritable key, Text value, Context context) throws IOException, InterruptedException {
20             String line = value.toString();
21             StringTokenizer tokenizer = new StringTokenizer(line);
22             while (tokenizer.hasMoreTokens()) {
23                 word.set(tokenizer.nextToken());
24                 context.write(word, one);
25             }
26         }
27     }
28 }

```

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Rs. 155.00

Elastic Search, Logstash, Kibana

ELK : Elastic Search

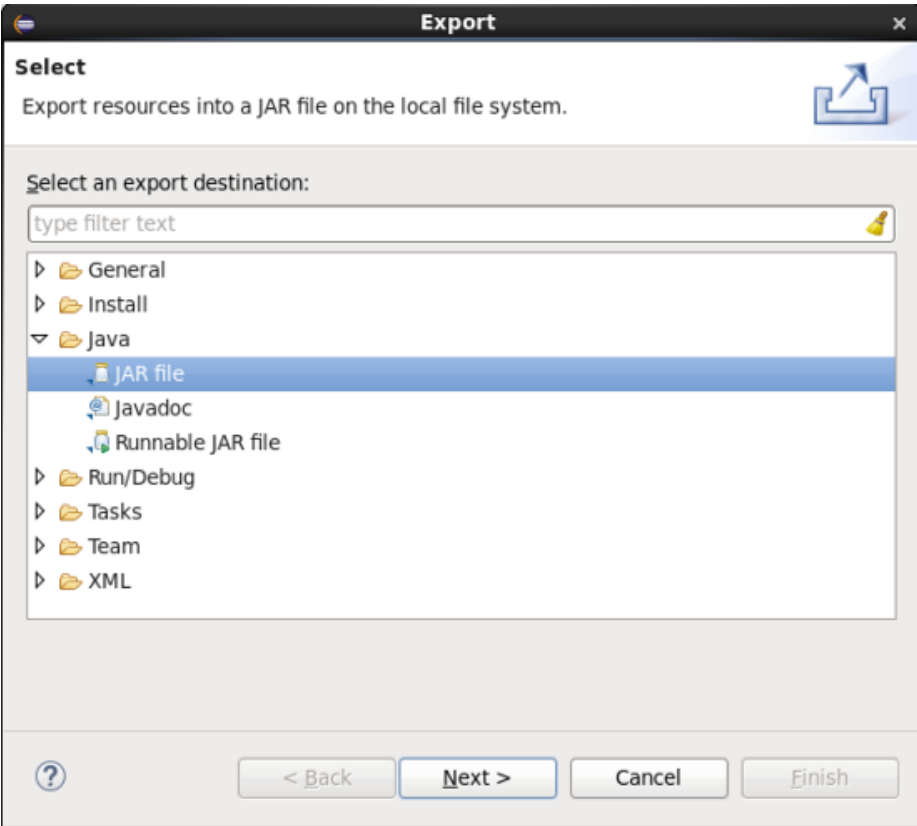
ELK : Logstash
with Elastic Search

ELK : Logstash,
ElasticSearch, and
Kibana 4

Exporting the Jar

Now we want to export it as a jar.

Right click on WordCount project and select "Export...":



DevOps

Phases of
Continuous
Integration

Software
development
methodology

DevOps

Examples of
continuous
integration (CI) /
continuous
delivery (CD) - Use
cases

Artifact repository
and repository
management

DevOps / Sys
Admin Q & A (I) -
Linux Commands

DevOps / Sys
Admin Q & A (II) -
Networks

DevOps / Sys
Admin Q & A (III) -
Linux Systems

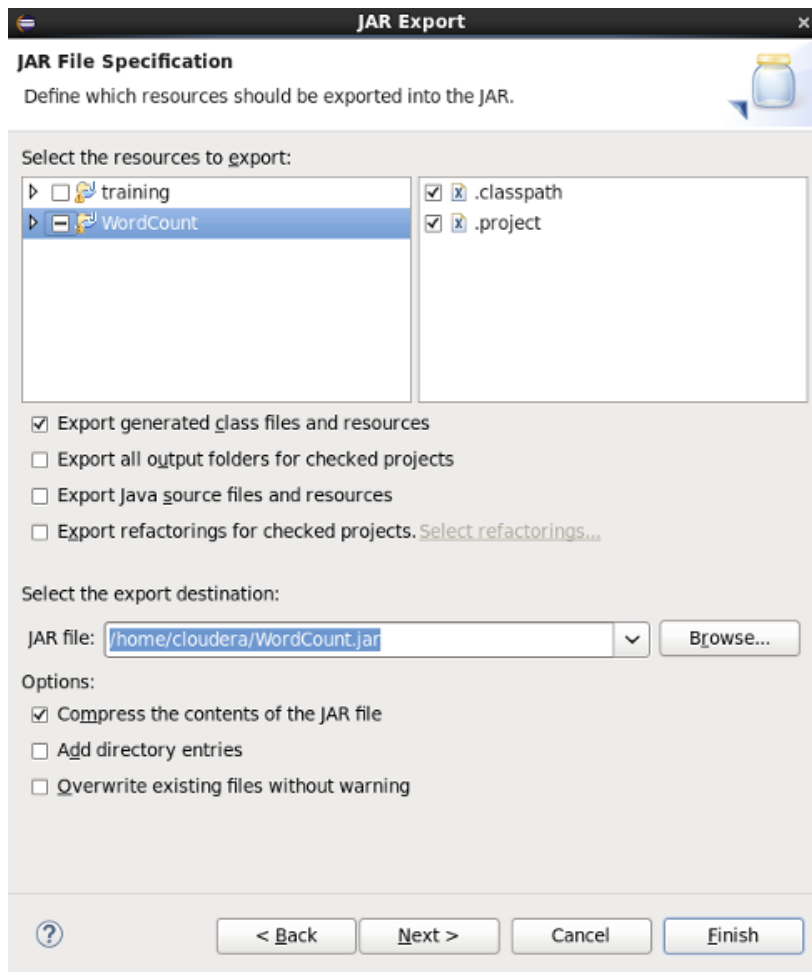
Nagios - The
industry standard
in IT infrastructure
monitoring on
Ubuntu

Jenkins

Install

Configuration -
Manage Jenkins -
security setup

Adding job and
build



[Scheduling jobs](#)

[Managing plugins](#)

[Git/GitHub plugins, SSH keys configuration, and Fork/Clone](#)

[JDK & Maven setup](#)

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[Build Action for GitHub Java application with Maven - Console Output, Updating Maven](#)

[Commit to changes to GitHub & new test results - Build Failure](#)

[Commit to changes to GitHub & new test results - Successful Build](#)

[Adding code coverage and metrics](#)

[Jenkins on EC2 - creating an EC2 account, ssh to EC2. and install](#)

[Jenkins on EC2 - account, plugins, and Configure System \(JAVA HOME\)](#)

[Jenkins on EC2 - Jenkinsfile, build, and Configure System](#)

[Jenkins on EC2 - Configuring GitHub Hook and Notification](#)

Input file

Here is our input:

```
[cloudera@quickstart ~]$ cat /home/cloudera/wordcount.txt
Hello Hadoop, Goodbye Hadoop.
```

Let's look at Hadoop file system:

```
[cloudera@quickstart ~]$ hadoop fs -ls
Found 3 items
drwx----- - cloudera cloudera      0 2015-03-24 09:41 .Trash
drwx----- - cloudera cloudera      0 2015-03-24 19:01 .staging
drwxr-xr-x - cloudera cloudera      0 2015-03-24 19:01 oozie-oozi
```

Now we want to copy the input file to hdfs:

```
[cloudera@quickstart ~]$ hadoop fs -mkdir input
[cloudera@quickstart ~]$ hadoop fs -put /home/cloudera/wordcount.txt input,
[cloudera@quickstart ~]$ hadoop fs -ls input
Found 1 items
-rw-r--r-- 1 cloudera cloudera      30 2015-03-24 20:23 input/wordcount.txt
[cloudera@quickstart ~]$ hadoop fs -cat input/wordcount.txt
Hello Hadoop, Goodbye Hadoop.
```

[cloudera@quickstart ~]\$

Pipeline &
Dependency
Graph Plugins

Jenkins Build Flow
Plugin

RUN

Time to run MapReduce job:

Puppet

```
[cloudera@quickstart ~]$ hadoop jar /home/cloudera/WordCount.jar WordCount
15/03/24 20:27:16 INFO client.RMPProxy: Connecting to ResourceManager at qui
15/03/24 20:27:16 INFO client.RMPProxy: Connecting to ResourceManager at qui
15/03/24 20:27:17 INFO mapred.FileInputFormat: Total input paths to process
15/03/24 20:27:17 INFO mapreduce.JobSubmitter: number of splits:2
15/03/24 20:27:17 INFO mapreduce.JobSubmitter: Submitting tokens for job: :
15/03/24 20:27:17 INFO impl.YarnClientImpl: Submitted application applicati
15/03/24 20:27:17 INFO mapreduce.Job: The url to track the job: http://quic
15/03/24 20:27:17 INFO mapreduce.Job: Running job: job_1426957621065_0026
15/03/24 20:27:31 INFO mapreduce.Job: Job job_1426957621065_0026 running in
15/03/24 20:27:31 INFO mapreduce.Job: map 0% reduce 0%
15/03/24 20:27:45 INFO mapreduce.Job: map 100% reduce 0%
15/03/24 20:27:57 INFO mapreduce.Job: map 100% reduce 100%
15/03/24 20:27:58 INFO mapreduce.Job: Job job_1426957621065_0026 completed
15/03/24 20:27:58 INFO mapreduce.Job: Counters: 50
```

File System Counters

```
FILE: Number of bytes read=64
FILE: Number of bytes written=327115
FILE: Number of read operations=0
FILE: Number of large read operations=0
FILE: Number of write operations=0
HDFS: Number of bytes read=279
HDFS: Number of bytes written=38
HDFS: Number of read operations=9
HDFS: Number of large read operations=0
HDFS: Number of write operations=2
```

Job Counters

```
Launched map tasks=2
Launched reduce tasks=1
Data-local map tasks=2
Total time spent by all maps in occupied slots (ms)=25340
Total time spent by all reduces in occupied slots (ms)=6288
Total time spent by all map tasks (ms)=25340
Total time spent by all reduce tasks (ms)=6288
Total vcore-seconds taken by all map tasks=25340
Total vcore-seconds taken by all reduce tasks=6288
Total megabyte-seconds taken by all map tasks=25948160
Total megabyte-seconds taken by all reduce tasks=6438912
```

Map-Reduce Framework

```
Map input records=1
Map output records=4
Map output bytes=46
Map output materialized bytes=76
Input split bytes=234
Combine input records=4
Combine output records=4
Reduce input groups=4
Reduce shuffle bytes=76
Reduce input records=4
Reduce output records=4
Spilled Records=8
Shuffled Maps =2
Failed Shuffles=0
Merged Map outputs=2
GC time elapsed (ms)=448
CPU time spent (ms)=1380
```

```
Physical memory (bytes) snapshot=617631744
Virtual memory (bytes) snapshot=2567532544
Total committed heap usage (bytes)=482017280

Shuffle Errors
BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0
WordCount$Map$Counters
INPUT_WORDS=4
File Input Format Counters
Bytes Read=45
File Output Format Counters
Bytes Written=38
[cloudera@quickstart ~]$
```

output

Here is our output from the MR run:

```
[cloudera@quickstart ~]$ hadoop fs -ls output
Found 2 items
-rw-r--r-- 1 cloudera cloudera 0 2015-03-24 20:27
output/_SUCCESS
-rw-r--r-- 1 cloudera cloudera 38
2015-03-24 20:27 output/part-00000
[cloudera@quickstart ~]$ hadoop fs -cat output/part-00000
Goodbye 1 Hadoop, 1 Hadoop. 1 Hello 1
```

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[Design Patterns \(C++\): Introduction - 2015](#)

[Hadoop 2.4 - Running a MapReduce Job - 2015](#)

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[Puppet Express 2](#)

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Chef

What is Chef?

Chef install on
Ubuntu 14.04 -
Local Workstation
via omnibus
installer

Setting up Hosted
Chef server

VirtualBox via
Vagrant with Chef
client provision

Creating and using
cookbooks on a
VirtualBox node

Chef server install
on Ubuntu 14.04

Chef workstation
setup on EC2
Ubuntu 14.04

Chef Client Node -
Knife
Bootstrapping a
node on EC2
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Docker 1.7

Docker install on
Amazon Linux AMI

Docker install on
EC2 Ubuntu 14.04

Docker container
vs Virtual Machine

Docker install on
Ubuntu 14.04

Docker Hello
World Application

Docker image and
container via
docker commands
(search, pull, run,
ps, restart, attach,
and rm)

More on docker
run command
(docker run -it,

docker run --rm,
etc.)

File sharing
between host and
container (docker
run -d -p -v)

Linking containers
and volume for
datastore

Dockerfile - Build
Docker images
automatically I -
FROM,
MAINTAINER, and
build context

Dockerfile - Build
Docker images
automatically II -
revisiting FROM,
MAINTAINER, build
context, and
caching

Dockerfile - Build
Docker images
automatically III -
RUN

Dockerfile - Build
Docker images
automatically IV -
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Dockerfile - Build
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WORKDIR, ENV,
ADD, and
ENTRYPOINT

Installing LAMP via
puppet on Docker

Docker install via
Puppet

Vagrant

Vagrant &
VirtualBox install
on Ubuntu 14.04

Creating a
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Provisioning

Networking - Port
Forwarding

Vagrant Share

Vagrant Rebuild &
Teardown

AWS (Amazon Web Services)

AWS : Creating a
snapshot (cloning
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AWS : Attaching
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AWS : Creating an
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AWS : S3 (Simple
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AWS : S3 (Simple
Storage Service) 2 -
Creating and
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AWS : S3 (Simple
Storage Service) 3 -
Bucket Versioning

AWS : S3 (Simple
Storage Service) 4 -
Uploading a large
file

AWS : S3 (Simple
Storage Service) 5 -
Uploading
folders/files
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AWS : S3 (Simple
Storage Service) 6 -
Bucket Policy for
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View/Download

AWS : S3 (Simple
Storage Service) 7 -
How to Copy or
Move Objects
from one region to
another

AWS : S3 (Simple
Storage Service) 8 -
Archiving S3 Data
to Glacier

AWS : Creating a CloudFront distribution with an Amazon S3 origin

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AWS : VirtualBox on EC2

AWS : NTP setup on EC2

AWS & OpenSSL : Creating / Installing a Server SSL Certificate

AWS : OpenVPN Access Server 2 Install

AWS : VPC (Virtual Private Cloud) 1 - netmask, subnets, default gateway, and CIDR

AWS : VPC (Virtual Private Cloud) 2 - VPC Wizard

AWS : VPC (Virtual Private Cloud) 3 - VPC Wizard with NAT

AWS - OpenVPN Protocols : PPTP, L2TP/IPsec, and OpenVPN

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AWS : Detecting stopped instance and sending an alert email using Mandrill smtp

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AWS : Redshift data warehouse
AWS : RDS Connecting to a DB Instance Running the SQL Server Database Engine
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AWS : RDS PostgreSQL 2 - Creating/Deleting a Table
AWS : MySQL Replication : Master-slave
AWS RDS : Cross-Region Read Replicas for MySQL and Snapshots for PostgreSQL
AWS : Restoring Postgres on EC2 instance from S3 backup

Redis

Redis vs Memcached
Redis 3.0.1 Install

Powershell 4 Tutorial

Powersehll :

Introduction
Powersehll : Help System
Powersehll : Running commands
Powersehll : Providers
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Windows Management Instrumentation (WMI)
How to Enable Multiple RDP Sessions in Windows 2012 Server
How to install and configure FTP server on IIS 8 in Windows 2012 Server
How to Run Exe as a Service on Windows 2012 Server
SQL Inner, Left, Right, and Outer Joins

Git/GitHub Tutorial

One page express tutorial for GIT and GitHub

Installation
add/status/log
commit and diff
git commit --amend

Deleting and
Renaming files

Undoing Things :
File Checkout &
Unstaging

Reverting commit

Soft Reset - (git
reset --soft <SHA
key>)

Mixed Reset -
Default

Hard Reset - (git
reset --hard <SHA
key>)

Creating &
switching
Branches

Fast-forward
merge

Rebase & Three-
way merge

Merge conflicts
with a simple
example

GitHub Account
and SSH

Uploading to
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GUI

Branching &
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Merging conflicts

GIT on Ubuntu and
OS X - Focused on
Branching

Setting up a
remote repository
/ pushing local
project and
cloning the remote
repo

Fork vs Clone,
Origin vs
Upstream

Git/GitHub
Terminologies

Git/GitHub via
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Branching &
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Git/GitHub via
SourceTree III : Git
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Git/GitHub via
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Subversion

Subversion Install
On Ubuntu 14.04

Subversion
creating and
accessing I

Subversion
creating and
accessing II
